



LOVELY
PROFESSIONAL
UNIVERSITY

Project Report

On

A CAMPUSBITE PRE-FOOD ORDERING PLATFORM

Submitted to

LOVELY PROFESSIONAL UNIVERSITY

In partial fulfilment of the requirements for the award of degree of

Bachelor of Computer Applications (BCA)

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CERTIFICATE

This is to certify that **Adhithyan V.V. (12322517), Bommanaboina Srinivasulu (12320599), Abhinav T.S. (12306552), Michael Gatsi (12302727), and Nadil (12303534)** have successfully completed the **pre-project planning** titled **“CampusBite”** under my supervision and guidance.

To the best of my knowledge and belief, this work represents their independent effort and original study. This report has not been submitted, either in whole or in part, to any other university or institution for the award of any degree or diploma.

The pre-project report is therefore approved and recommended for submission as partial fulfilment of the requirements for the award of the Bachelor of Computer Application degree.

Declaration

We hereby declare that the project entitled “**CampusBite — Web-Based Campus Food Pre-Ordering System**” is our original work carried out as part of the requirements for the degree of Bachelor of Computer Applications. The work presented in this report has been completed under the supervision of our project mentor and has not been submitted previously for any other academic qualification.

Acknowledgement

The successful completion of this project would not have been possible without the guidance and support of many individuals. We would like to express our sincere gratitude to our project supervisor for their valuable guidance, suggestions, and encouragement throughout the project development process.

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Chapter 1 – Introduction

With the rapid development of digital technologies, many everyday services are transitioning from traditional manual processes to online systems. Web-based applications have become essential tools for improving efficiency, convenience, and accessibility in various sectors including education, commerce, and service industries.

Food courts in universities often experience heavy congestion during peak hours such as lunch breaks. Students usually spend considerable time standing in queues to place orders and collect food. This results in wasted time, overcrowding, and inefficiency for both students and food stall vendors.

To address this problem, the **CampusBite platform** has been proposed. CampusBite is a web-based food pre-ordering system designed specifically for college campuses. The system allows students to browse food stalls, view menus, and place orders online before arriving at the stall.

After an order is placed and payment is completed, the system generates a unique order token. This token enables students to collect their food directly from the vendor without waiting in long queues.

The platform uses a role-based architecture that includes three types of users:

- **Administrator** – manages the entire platform
- **Restaurant Owner / Vendor** – manages stalls and menus
- **User / Student** – browses menus and places orders

CampusBite aims to enhance convenience for students and improve order management efficiency for food stall vendor

Chapter 2 – Literature Review

Digital ordering systems have become increasingly popular in recent years due to the growth of online commerce and mobile technologies. Many restaurants and food delivery services use online platforms to allow customers to place orders remotely.

Research in service management shows that digital ordering platforms can significantly reduce waiting time and improve operational efficiency. These systems automate order processing, improve communication between customers and vendors, and help businesses manage demand more effectively.

Several studies have also explored queue management techniques in service environments. Long queues are often caused by inefficient ordering processes and limited-service capacity. Implementing online ordering systems allows customers to place orders remotely, thereby reducing congestion at service points.

Role-based web applications have also been widely adopted in modern information systems. These systems assign different permissions to different types of users, ensuring that each user can only access relevant functionalities.

Despite the availability of online ordering systems, most existing platforms are designed for large commercial restaurants rather than campus food courts. Therefore, there is a need for a specialized system that addresses the unique needs of university environments.

Chapter 3 – Problem Statement and Scope

Problem Statement

Students in university food courts frequently experience long waiting times due to manual ordering processes. During peak hours, the large number of students ordering food simultaneously leads to overcrowding and inefficient service.

The lack of a digital ordering system prevents students from placing orders in advance, resulting in unnecessary delays.

Scope of the Project

The CampusBite platform focuses on providing a digital solution for campus food courts. The system enables:

- Online food ordering for students
- Menu and stall management for vendors
- Order tracking using unique tokens
- Platform monitoring by administrators

The system is intended to operate within university campuses and can be expanded to other institutional environments

Chapter 4 – Objectives of the Study

The primary objectives of the CampusBite system are:

1. To develop a web-based food ordering platform for university campuses.
2. To reduce waiting time in food courts by enabling pre-ordering.
3. To allow vendors to manage food stalls and menu items efficiently.
4. To generate unique tokens for order identification and pickup.
5. To provide role-based access control for administrators, vendors, and users.
6. To improve order management and service efficiency within campus food courts

Chapter 5 – Research Methodology

The development of the CampusBite system followed a structured methodology consisting of several stages.

Requirement Analysis

Information was gathered through discussions with students and vendors to identify common problems faced during food ordering in campus environments.

System Design

The architecture and workflow of the system were designed using diagrams such as Data Flow Diagrams and system architecture models.

Development

The system was implemented using web technologies including HTML, CSS, JavaScript, PYTHON,DJANGO and POSTGRESQL.

Testing

The application was tested using various testing techniques to ensure correct functionality and reliabilit

Chapter 6 — System Analysis

The traditional food ordering process requires students to stand in queues and wait for their orders. This process becomes inefficient when many students attempt to order food at the same time.

The proposed system improves this process by introducing an online ordering platform where students can browse menus and place orders in advance.

Advantages of the proposed system include:

- Reduced waiting time
- Improved order management
- Better communication between vendors and customers
- Increased convenience for user

Chapter 7 – Data Flow Diagrams

Data Flow Diagrams represent the flow of information within the system.

Level 0 DFD

Entities involved:

- Student
- Vendor
- Administrator

The system receives user input, processes orders, and returns order tokens.

Level 1 DFD

Processes include:

- User registration and login
- Menu browsing
- Order placement
- Payment processing
- Order management

Chapter 8 — Requirement Specification

Hardware Requirements

- Personal computer or laptop
- Internet connection
- Server environment

Software Requirements

- Web browser
- PYTHON programming language
- POSTGRESQL database
- HTML, CSS, JavaScript
- Visual Studio Code or similar ID

Chapter G — System Design and Architecture

The CampusBite system follows a client-server architecture.

System Components

Client (User
Interface) Web
Server Database
Server

User Roles

Administrator
Restaurant Owner
Student/User

Database Tables

Users
Restaurants
Menu_Items
Orders
Order_Details
Payments

Chapter 10 – Implementation

The system consists of several modules.

User Module

Students can create accounts, browse food stalls, and place orders.

Vendor Module

Vendors can manage their stalls, update menus, and process incoming orders.

Administrator Module

The administrator monitors platform activities and manages user accounts.

Token Generation Module

Each successful order generates a unique token that allows students to collect their food quickly.

Chapter 11 – Testing

Two primary testing methods were used.

Black Box Testing

The system was tested based on input and output without considering internal code structure.

Example test cases:

- User login validation
- Menu display functionality
- Order placement process

White Box Testing

Internal program logic was tested to ensure proper execution of system components

Chapter 12 – Security Measures

Several security mechanisms were implemented.

Secure Login

Users must authenticate using valid credentials.

Data Encryption

Sensitive data such as payment information is protected.

Role-Based Access Control

Different users have different permissions within the system.

Secure Server Environment

The system runs on a protected server environment to prevent unauthorized access

Chapter 13 – Future Scope

Future improvements may include:

- Mobile application development
- QR-code-based food pickup
- AI-based demand prediction
- Recommendation systems for food items
- Advanced analytics for vendors

Chapter 14 – Conclusion

The CampusBite system provides an efficient solution for managing food ordering within university campuses. By enabling students to pre-order food online, the system reduces waiting times and improves the overall dining experience.

The platform also helps vendors manage orders more effectively and maintain better control over their menu offerings. Through its role-based design and web-based interface, the system ensures accessibility and scalability.

Overall, CampusBite demonstrates how digital technologies can enhance service efficiency and convenience in campus environment

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